REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 1

Fundamentals of Orbital Mechanics

The Two-Body Problem, Keplerian Motion, and Orbital Elements

Prepared for	Cao Khanh Tan
Module	1 of the Foundation Phase
Programme	4 months | 12 to 15 hours per week
Tools introduced	GMAT, STK, MATLAB / Simulink, Python (poliastro, astropy)

SESSION 1

Fundamentals of Orbital Mechanics

This is the first content module of the Astrodynamics Specialist programme and the foundation for everything that follows. Orbit transfers, perturbations, constellation design and interplanetary trajectories are all built on the three ideas covered here: the two-body problem, Keplerian motion, and the orbital elements. Work through the sections in order, keep a notebook for the derivations, and attempt the practice set at the end before our next live session.

Learning objectives

By the end of this module you should be able to:

- State the assumptions of the two-body problem and write its equation of motion.
- Explain why every unperturbed orbit is a conic section and identify the four conic types by eccentricity.
- Apply Kepler three laws and the vis-viva equation to compute orbital speed, radius and period.
- Convert between the three anomalies and solve Kepler equation numerically.
- Describe the six classical orbital elements and what each one controls geometrically.
- Reproduce the worked examples using Python (poliastro / astropy) and sketch the same setup in STK.

Where this fits in the programme

The programme moves through four phases: a Foundation Phase (this module), a Core Training Phase (perturbations, transfer orbits, constellation design), an Advanced Application Phase (station-keeping, rendezvous, GNSS, re-entry) and a Capstone. The competencies you build now are used directly in the later practical work, including the Earth to Mars transfer, the satellite-constellation design and the orbit-maintenance simulations listed on the programme page.

Notation and reference constants

Symbol / constant	Meaning	Value
mu (Earth)	Standard gravitational parameter G M	398,600 km^3 / s^2
R (Earth, equatorial)	Mean equatorial radius	6,378 km
mu (Sun)	Solar gravitational parameter	1.327 x 10^11 km^3 / s^2
1 AU	Astronomical unit	1.496 x 10^8 km
Sidereal day	Earth rotation period (inertial)	86,164 s
a, e, i	Semi-major axis, eccentricity, inclination	see Section on elements
Omega, omega, nu	RAAN, argument of periapsis, true anomaly	angles, in degrees or radians

Throughout, we work in kilometres and seconds, so speeds come out in km/s. Vectors are shown in bold; their magnitudes are the same letter in plain type.

PART 1

The Two-Body Problem

Astrodynamics begins with two point masses attracting each other through gravity and nothing else. Newton universal law of gravitation gives the force between a body of mass m1 and a body of mass m2 separated by a distance r:

$$F = G\frac{m_1 m_2}{r^2}$$

For a satellite orbiting the Earth, m2 (the satellite) is negligible next to m1 (the Earth), so we track the satellite position vector r relative to the Earth centre. Combining the two equations of motion and reducing to relative motion gives the single governing equation of the two-body problem:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}$$

where mu is the standard gravitational parameter, a single number that bundles G and the central mass together:

$$\mu = G(m_1 + m_2) \approx GM_\oplus$$

> **Why mu instead of G and M separately**
> mu is known far more precisely than G or M individually, because it is what we actually measure from spacecraft tracking. For Earth, mu = 398,600 km^3 / s^2. Memorise this number, you will use it constantly.

Assumptions (and when they break)

The clean two-body result relies on four idealisations:

- The two bodies are point masses, or perfectly spherical with uniform density.
- Gravity is the only force acting. No atmospheric drag, no thrust.
- No third body. The Sun and Moon are ignored.
- The central body does not radiate pressure or have an irregular shape.

Real orbits violate all four to some degree. The Earth is oblate (the J2 effect), the upper atmosphere drags low orbits, the Sun and Moon tug on high orbits, and solar radiation pushes on large surfaces. These are the perturbations we study in the Core Phase. For now, the two-body model is accurate enough to design and understand an orbit, and it is the reference against which every perturbation is measured.

Three quantities that never change

The power of the two-body problem comes from what it conserves. From the equation of motion you can prove that three quantities stay constant along the entire orbit.

1. Specific angular momentum. The cross product of position and velocity is fixed in both size and direction:

$$\mathbf{h} = \mathbf{r} \times \mathbf{v} = \text{const}$$

Because its direction is fixed, the motion stays in a single plane: the orbital plane. Its magnitude also gives Kepler second law directly.

2. Specific mechanical energy. Kinetic plus potential energy per unit mass is constant:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = \text{const}$$

3. The eccentricity vector. This one points from the focus toward periapsis and its magnitude is the eccentricity e:

$$\mathbf{e} = \frac{1}{\mu}\left[\left(v^2 - \frac{\mu}{r}\right)\mathbf{r} - (\mathbf{r} \cdot \mathbf{v})\,\mathbf{v}\right]$$

Key idea
Conservation of angular momentum flattens the problem to a plane, and conservation of energy ties speed to distance. Together they force the orbit to be a conic section. Almost everything in this module is a consequence of these two facts.

PART 2

Conic Sections and the Orbit Equation

Solving the two-body equation of motion for the shape of the path (rather than position versus time) yields the orbit equation, one of the most important results in the field:

$$r = \frac{p}{1 + e\cos\nu}, \qquad p = \frac{h^2}{\mu}$$

Here nu is the true anomaly, the angle at the focus measured from periapsis to the current position, and p is the semi-latus rectum. This is the polar equation of a conic section with the focus at the central body. The eccentricity e alone decides which conic you get:

Eccentricity	Conic	Energy	Bound?
e = 0	Circle	Negative	Yes, closed
0 < e < 1	Ellipse	Negative	Yes, closed
e = 1	Parabola	Zero	No, escape
e > 1	Hyperbola	Positive	No, flyby / escape

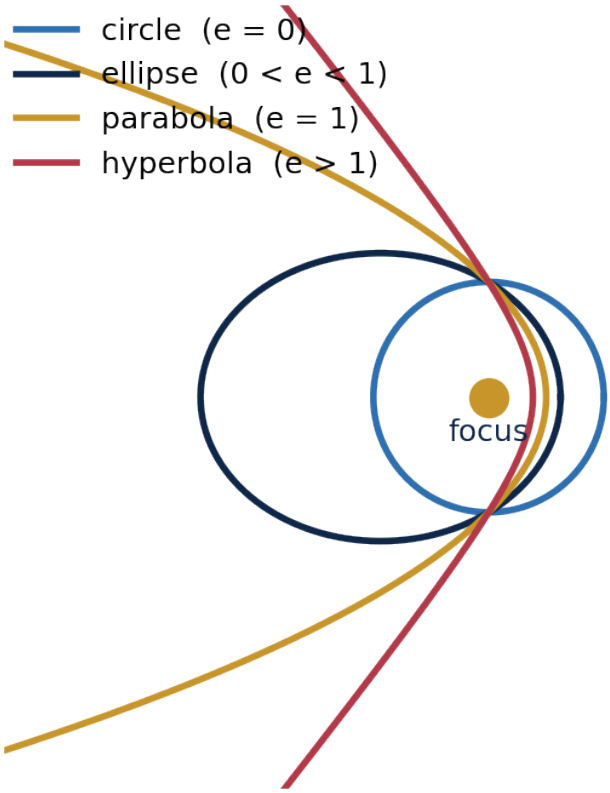


Figure 1. All four conics drawn with a common focus and the same semi-latus rectum p. Increasing eccentricity opens the curve until, at e = 1, it stops closing and the orbit becomes an escape trajectory.

Geometry of the ellipse

Closed orbits are ellipses, so it is worth knowing the ellipse cold. The nearest point to the focus is periapsis, the farthest is apoapsis (perigee and apogee for Earth orbits). The key relations are:

$$r_p = a(1-e), \quad r_a = a(1+e), \quad b = a\sqrt{1-e^2}, \quad p = a(1-e^2)$$

$$a = \frac{r_p + r_a}{2}, \qquad e = \frac{r_a - r_p}{r_a + r_p}$$

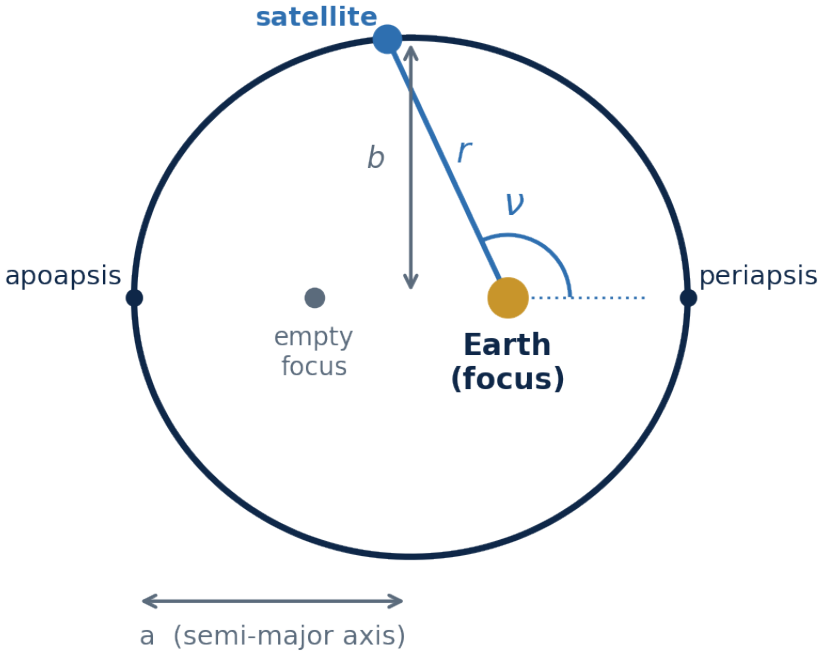


Figure 2. Orbit geometry. The Earth sits at one focus, not at the centre. The satellite position is set by the radius r and the true anomaly nu measured from periapsis. The semi-major axis a fixes the size of the orbit and, as we will see, its energy and period.

PART 3

Keplerian Motion

Kepler discovered his three laws empirically from planetary data; Newton later showed they follow from the two-body problem. They remain the language of orbit design.

Kepler three laws

First law. Each orbit is an ellipse with the central body at one focus.

Second law. The line from the central body to the satellite sweeps equal areas in equal times. This is just conservation of angular momentum: a satellite moves fastest at periapsis and slowest at apoapsis.

Third law. The square of the period is proportional to the cube of the semi-major axis.

The vis-viva equation

The single most useful formula in this course comes from combining the energy integral with the geometry of the ellipse. It gives the speed at any point in the orbit from just the current radius and the semi-major axis:

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$$

Its companion links the energy directly to the semi-major axis, which is why a alone determines the size and period of an orbit:

$$\varepsilon = -\frac{\mu}{2a}$$

How to read vis-viva

For a circular orbit r = a, so v = sqrt(mu / r): a single speed. For an ellipse, plug in r = r_p to get the maximum speed at periapsis and r = r_a for the minimum at apoapsis. For a parabola a is infinite, so v = sqrt(2 mu / r): the local escape speed.

Period and mean motion

Kepler third law, written with mu, gives the orbital period and the mean motion n (the average angular rate):

$$T = 2\pi\sqrt{\frac{a^3}{\mu}}$$

$$n = \sqrt{\frac{\mu}{a^3}} = \frac{2\pi}{T}$$

Note what is absent: the period depends only on a and mu, not on the eccentricity. Two orbits with the same semi-major axis but very different shapes complete one revolution in exactly the same time.

PART 4

Time and the Three Anomalies

The orbit equation tells you the shape of the path but not where the satellite is at a given time. Relating position to time is the classic "time of flight" problem, and it needs three related angles.

- **True anomaly (nu)**: the real geometric angle from periapsis to the satellite, seen from the focus.
- **Eccentric anomaly (E)**: an auxiliary angle defined on the circle that circumscribes the ellipse.
- **Mean anomaly (M)**: a fictitious angle that grows linearly with time at the mean motion n.

Mean anomaly is the easy one, because it is linear in time:

$$M = n\,(t - t_p)$$

Mean anomaly connects to eccentric anomaly through Kepler equation, the famous transcendental relation:

$$M = E - e\sin E$$

and eccentric anomaly connects to the true anomaly (and to the radius) by:

$$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\,\tan\frac{E}{2}$$

$$r = a\,(1 - e\cos E)$$

Solving Kepler equation

Because M = E - e sin E cannot be inverted in closed form, we solve for E numerically. Newton-Raphson converges in a handful of iterations for typical eccentricities, starting from E0 = M:

$$E_{k+1} = E_k - \frac{E_k - e\sin E_k - M}{1 - e\cos E_k}$$

> **The forward and inverse problems**
> Going from time to position (given M, find E, then nu) needs the iteration above. Going from position to time (given nu, find E, then M, then t) is direct, no iteration required. You will implement both in code this week.

A related quantity worth knowing is the flight path angle gamma, the angle between the velocity vector and the local horizontal. It is zero at periapsis and apoapsis and maximal in between:

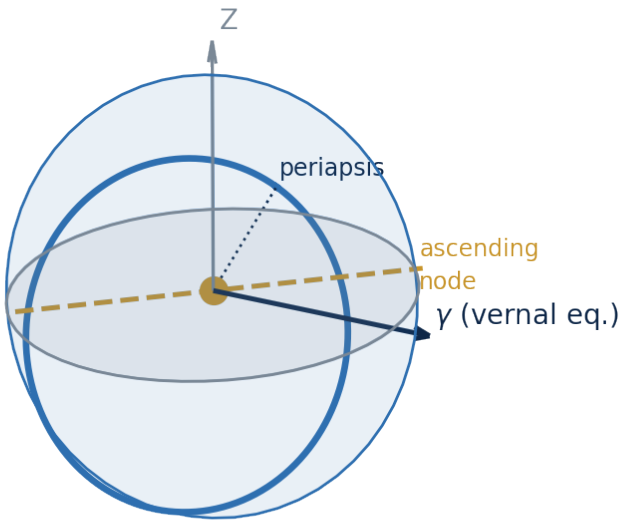
$$\tan\gamma = \frac{e\sin\nu}{1 + e\cos\nu}$$

PART 5

The Six Classical Orbital Elements

A position and velocity vector fully describe an orbit, but they are hard to interpret. The classical (Keplerian) elements describe the same orbit with six numbers that each have a clear geometric meaning: two set the size and shape, three orient the orbit in space, and one locates the satellite along it.

Element	Symbol	Controls
Semi-major axis	**a**	Size of the orbit (and therefore energy and period)
Eccentricity	**e**	Shape, from circular (0) to increasingly elongated
Inclination	**i**	Tilt of the orbital plane relative to the reference (equatorial) plane
RAAN	**Omega**	Swivel of the orbital plane about the polar axis, measured from the vernal equinox
Argument of periapsis	**omega**	Orientation of the ellipse within the orbital plane
True anomaly	**nu**	Position of the satellite along the orbit at a given epoch



i: inclination Ω: RAAN *ω*: argument of periapsis *ν*: true anomaly

Figure 3. The three orientation angles. The orbital plane (blue) is tilted from the equatorial plane (grey) by the inclination i. The two planes meet along the line of nodes; the ascending node is where the satellite crosses going north. RAAN (Omega) is measured in the equatorial plane from the vernal equinox to that node, and the argument of periapsis (omega) is measured in the orbital plane from the node to periapsis.

Instead of true anomaly
The sixth element is often given as the mean anomaly M or the time of periapsis passage rather than nu, because M is what propagates linearly in time. All three carry the same information once you can solve Kepler equation.

Reference frames in one paragraph

Elements are only meaningful against a frame. The Earth-Centred Inertial frame (ECI, for example J2000) is fixed relative to the stars and is where orbits are naturally described. The Earth-Centred Earth-Fixed frame (ECEF) rotates with the planet and is used for ground tracks and geographic coverage. The perifocal frame (PQW) sits in the orbital plane with its axis pointing at periapsis, which is where the orbit equation is simplest. Converting between these frames is a routine but essential skill, and STK and poliastro both handle it for you once you know what each frame means.

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Common Orbit Families

You will meet these repeatedly. The point is not to memorise numbers but to connect each orbit to the elements that define it.

Orbit	Typical altitude	Defining feature	Used for
LEO (low Earth)	160 to 2,000 km	Short period, ~90 min	Imaging, ISS, large constellations
MEO (medium)	~20,000 km	Half-day period	GNSS: GPS, Galileo, BeiDou
GEO (geostationary)	35,786 km	24 h period, i near 0	Communications, weather
Sun-synchronous	600 to 800 km	Inclination near 98 deg, uses J2	Earth observation, consistent lighting
Molniya	Highly elliptical	e ~ 0.7, i = 63.4 deg	High-latitude coverage
GTO (transfer)	LEO to GEO ellipse	Parking ellipse before circularising	Reaching GEO

Worked Examples

Example 1. Period of a low circular orbit (ISS altitude, 420 km)

The orbit is circular so a = R + h = 6,378 + 420 = 6,798 km. Apply Kepler third law:

$$T = 2\pi\sqrt{a^3/\mu}$$

Substituting mu = 398,600 gives T = 5,578 s, which is 92.97 minutes. That is the familiar "the station orbits about every hour and a half". Vis-viva with r = a gives an orbital speed of 7.66 km/s.

Example 2. Circular speed at 400 km

For a circular orbit the vis-viva equation collapses to a single speed:

$$v_{\text{circ}} = \sqrt{\mu/r}$$

With r = 6,378 + 400 = 6,778 km, v = sqrt(398,600 / 6,778) = 7.67 km/s. Low orbits are fast: roughly 7.7 km/s, about Mach 23.

Example 3. Radius of a geostationary orbit

A geostationary satellite must match one sidereal day, T = 86,164 s. Invert Kepler third law to solve for the semi-major axis:

$$a = \left(\frac{\mu T^2}{4\pi^2}\right)^{1/3}$$

This gives a = 42,164 km, an altitude of 35,786 km above the equator, at which the orbital speed is 3.07 km/s. This is why every communications dish on the ground can point at a fixed spot in the sky.

PRACTICE

Computing With the Tools

Everything above can be reproduced in a few lines of Python. The snippet below builds an orbit from classical elements and reports its period and apses. Type it out yourself rather than copying, and change the numbers to match the worked examples.

```
from astropy import units as u
from poliastro.bodies import Earth
from poliastro.twobody import Orbit

# Build an orbit from the six classical elements
orb = Orbit.from_classical(
    Earth,
    a   = 6798 * u.km,     # semi-major axis
    ecc = 0.0  * u.one,    # eccentricity
    inc = 51.6 * u.deg,    # inclination
    raan= 40   * u.deg,    # RAAN
    argp= 0    * u.deg,    # argument of periapsis
    nu  = 0    * u.deg)    # true anomaly

print(orb.period.to(u.min))     # ~ 92.97 min
print(orb.r_p.to(u.km), orb.r_a.to(u.km))
print(orb.v)                    # velocity vector in km/s
```

Tooling note

poliastro is the library named in the syllabus and the one we use in these modules. If you install a very recent Python stack and hit dependency issues, the actively maintained fork hapsira offers a nearly identical API. Either is fine for this course. In STK, build the same orbit through the satellite Orbit Wizard using the Classical (Keplerian) element set, then confirm the period in the 2D and 3D views. GMAT will be introduced alongside the transfer work in the Core Phase.

This Week (aim for 12 to 15 hours)

Bring your work to the next live session. Show your reasoning, not just final numbers.

Assignment: Orbital Mechanics Fundamentals

- **Reading and notes (3 to 4 h).** Re-derive the vis-viva equation from the energy integral in your own hand. Write a one-page summary of the six elements in your own words.
- **Hand calculations (3 to 4 h).** For an orbit with r_p = 500 km altitude and r_a = 35,786 km altitude: find a, e, the period, and the speeds at periapsis and apoapsis. (This is a GTO, so keep the result for later.)
- **Coding (4 to 5 h).** Reproduce all three worked examples in poliastro or astropy. Then write a short function that solves Kepler equation with Newton-Raphson and returns nu given M and e. Verify it against M = 1.0 rad, e = 0.35 (you should get nu near 97.6 degrees).
- **STK familiarisation (2 h).** Create the ISS-altitude orbit from Example 1 in STK, view the ground track, and note how changing inclination changes the coverage. Take two screenshots for your log.

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One-Page Formula Sheet

Equation of motion	$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r}$
Orbit equation	$r = \frac{p}{1+e\cos\nu}, \qquad p = \frac{h^2}{\mu}$
Ellipse geometry	$r_p = a(1-e), \quad r_a = a(1+e), \quad b = a\sqrt{1-e^2}, \quad p = a(1-e^2)$
Vis-viva	$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)$
Energy and a	$\varepsilon = -\frac{\mu}{2a}$
Period	$T = 2\pi\sqrt{\frac{a^3}{\mu}}$
Mean motion	$n = \sqrt{\frac{\mu}{a^3}} = \frac{2\pi}{T}$
Kepler equation	$M = E - e\sin E$
Newton-Raphson step	$E_{k+1} = E_k - \frac{E_k - e\sin E_k - M}{1 - e\cos E_k}$
True vs eccentric anomaly	$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\,\tan\frac{E}{2}$

Key glossary

- **Periapsis / apoapsis:** closest / farthest points to the focus (perigee / apogee for Earth).
- **Semi-latus rectum p:** the radius at nu = 90 deg; p = a(1 - e^2).
- **Line of nodes:** the intersection of the orbital and equatorial planes.
- **Epoch:** the reference time at which the elements are stated.
- **Delta-v:** the change in velocity a manoeuvre requires; the currency of orbit transfers, coming next phase.

Going further

Three standard texts cover this material in depth if you want more: Curtis, Orbital Mechanics for Engineering Students; Vallado, Fundamentals of Astrodynamics and Applications; and Bate, Mueller and White, Fundamentals of Astrodynamics. These are real, widely used references, but I cannot verify exact editions or page numbers from here, so please confirm those details yourself before citing them.

Coming up in the Core Phase

With the fundamentals in place, the next modules add the forces the two-body model ignores (perturbations and station-keeping) and then put orbits to work with transfer manoeuvres: Hohmann, bi-elliptic and low-thrust. The delta-v and vis-viva tools from this week carry straight over.